The Design and Implementation of a Closed-Loop Filament Diameter Measurement and Control System for a 3D Printing Filament Extruder

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Abstract

This report documents an internal internship carried out at the German University in Cairo, in which a closed-loop system for measuring and controlling the diameter of 3D-printing filament was designed and built for an EX6 Filabot extruder. Instead of optical or laser sensors, the system uses a mechanical amplifier that couples the filament thickness to a magnet, whose position is sensed by a linear Hall-effect sensor. The analog signal is digitized by an ADS1115 converter, mapped to millimeters through a second-order polynomial calibration, and fed to a PID controller with dead-time compensation running on a Raspberry Pi 5, which regulates the speed of the pulling motor. The report describes the host organization, the hardware and software developed, the calibration and control methodology, and the professional skills gained during the internship.

Keywords: filament extrusion, closed-loop control, PID, Hall-effect sensor, Raspberry Pi, ADS1115, 3D printing

I. Introduction

The primary goal of this internship was to gain hands-on engineering experience by translating theoretical concepts from Electronics Engineering coursework into a practical solution. During the program, I studied 3D-printing filament production, focusing on material quality control with the EX6 Filabot extruder. The core objective was to design a dynamic, closed-loop control mechanism that tracks the filament diameter in real time and modulates the speed of the pulling motor to ensure uniform, high-quality extrusion.

The project integrated several disciplines within electronics engineering: embedded hardware, sensor interfacing, signal processing, and feedback control. Rather than using a standard optical sensor, the team engineered a measurement apparatus that combines a mechanical amplifier with a linear Hall-effect sensor, detecting minute variations in thickness through shifts in a magnetic field. A Raspberry Pi 5 processes the live readings and maps the analog signal to an exact diameter through a rigorous calibration. A PID controller then regulates the pulling motor automatically, correcting any dimensional discrepancy.

The remainder of this report is organized as follows. Section 2 describes the host organization. Section 3 presents the internship tasks in detail: the assigned project, the tools used, the measurement mechanism, the calibration and mathematical model, the closed-loop and PID control strategy, and the research carried out. Section 4 evaluates the internship experience, and Section 5 concludes the report.

II. Company / Organization Description

The German University in Cairo (GUC) is a private institution in Egypt that offers comprehensive higher education across a wide range of academic and engineering fields. The university is dedicated to advancing scientific research, fostering innovation, and equipping students with the hands-on skills needed for professional success. To support this mission, the GUC integrates practical learning opportunities, such as internships, laboratory research, and applied engineering projects, directly into its academic programs.

This internal internship took place within the GUC's Department of Design and Production Engineering, under the direct supervision of Dr. Amna Ramzy and Eng. Omar Abo Mandour. The department focuses on the intersection of manufacturing processes, mechanical design, production automation, and applied engineering problem-solving. Working in this multidisciplinary environment provided extensive exposure to the integration of electronic sensors, embedded systems, software programming, and mechanical control mechanisms.

The core of the internship was the engineering of a practical, closed-loop diameter measurement and control system for a 3D-printing filament extruder. The project was an ideal platform for bridging theoretical concepts from the Electronics Engineering curriculum and real-world industrial applications, and it allowed me to develop concrete skills in sensor calibration, precision measurement, PID control, and embedded software development.

III. Internship Performed Tasks

III.1. Assigned Internship Project

The primary technical objective assigned during the internship was the development of a complete closed-loop filament diameter measurement and control system for an EX6 Filabot 3D-printing extruder. The purpose of the system is to monitor the filament diameter continuously after cooling and to use this high-resolution data as real-time feedback to regulate the pulling mechanism. Conceptually, if the extruded material deviates from the target thickness, the speed of the pulling motor is adjusted to compensate.

The project required a thorough understanding of the extrusion process, together with the design of custom electronics that translate a mechanical displacement into a robust, quantifiable sensor signal. The overall system architecture can be summarized by the signal chain

$$\textit{Extrusion} \rightarrow \textit{Cooling} \rightarrow \textit{Measurement gauge} \rightarrow \textit{ADC conversion} \rightarrow \textit{Calibration} \rightarrow \textit{PID controller} \rightarrow \textit{Motor driver} \rightarrow \textit{Pulling motor} \rightarrow \textit{Filament diameter}$$

which establishes a continuous feedback loop in which the manufacturing output is monitored and used to determine the control action applied at the input.

III.2. Software and Engineering Tools

A diverse set of engineering tools and software environments was used to realize this electronics-focused project; they are summarized in Table 1.

The Raspberry Pi 5, programmed in Python through Thonny, served as the primary embedded controller. The Python application handles the backend PID computations and also provides a graphical user interface (GUI) built with tkinter, which lets operators enter target parameters and view live machine metrics, including the real-time diameter, the L298N-driven motor speed, and the raw sensor voltage. Autodesk Fusion was used for the CAD design of the custom measurement gauge, and Proteus was used for the schematic and the custom PCB layout of the sensor circuitry. I also gained extensive

experience with the I2C communication protocol required to transfer data from the ADS1115 ADC to the Raspberry Pi.

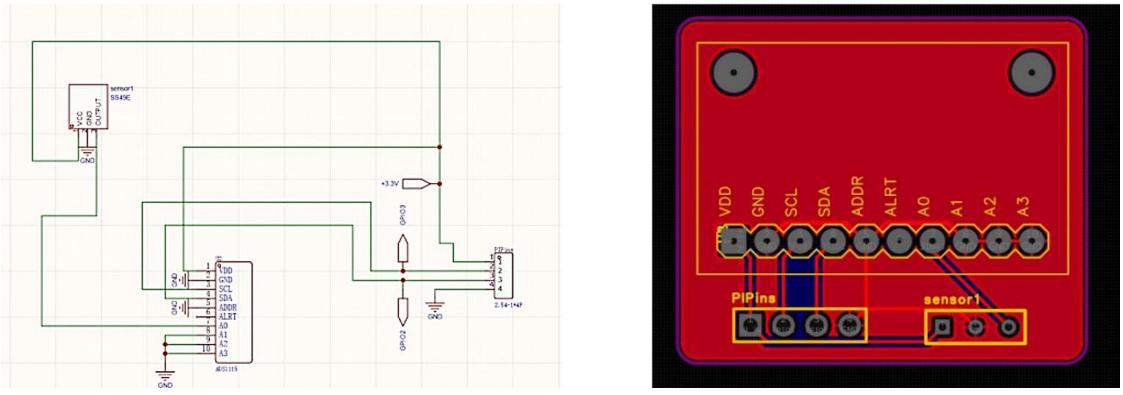
Table 1: Hardware and software used in the project and the role each one played.

Tool / Component	Role in the project
EX6 Filabot extruder	Produces the filament; the process is under control
SS49E linear Hall-effect sensor	Converts the magnet position into a continuous analog voltage
ADS1115 ADC	Digitizes the sensor voltage and sends it to the Pi over I2C
Raspberry Pi 5	Embedded controller: acquisition, calibration, PID, and GUI
L298N motor driver	Drives the pulling motor from the controller's PWM output
Python (Thonny, tkinter)	Control application and graphical user interface
Autodesk Fusion	CAD design of the custom measurement gauge
Proteus	Schematic capture and PCB layout of the sensor board
Microsoft Excel	Logging of calibration data and polynomial curve fitting
Laboratory 3D printers	Fabrication of the gauge housing

III.3. Filament Diameter Measurement Mechanism

A critical component of the project was the electromechanical apparatus that measures the filament diameter. To obtain a rigid and compact structure, a custom housing for the sensing and mechanical elements was designed in Autodesk Fusion and fabricated on the laboratory 3D printers. Inside this gauge, the filament passes between two bearings: one fixed, which guides the filament and acts as a stationary reference, and one movable, mounted on a sliding mechanism so that it follows the filament surface. Any fluctuation in the filament thickness therefore changes the relative position of the movable bearing. This displacement is transferred to a permanent magnet attached to the movable bearing, positioned next to a fixed analog SS49E Hall-effect sensor.

To secure these components and preserve signal integrity, a custom printed circuit board (PCB) was developed in Proteus. As shown in Figure 1, the board connects the analog Hall sensor to an ADS1115 analog-to-digital converter, which provides stable, high-resolution data transmission over the I2C protocol to the Raspberry Pi 5.



(a) Circuit schematic (b) PCB layout

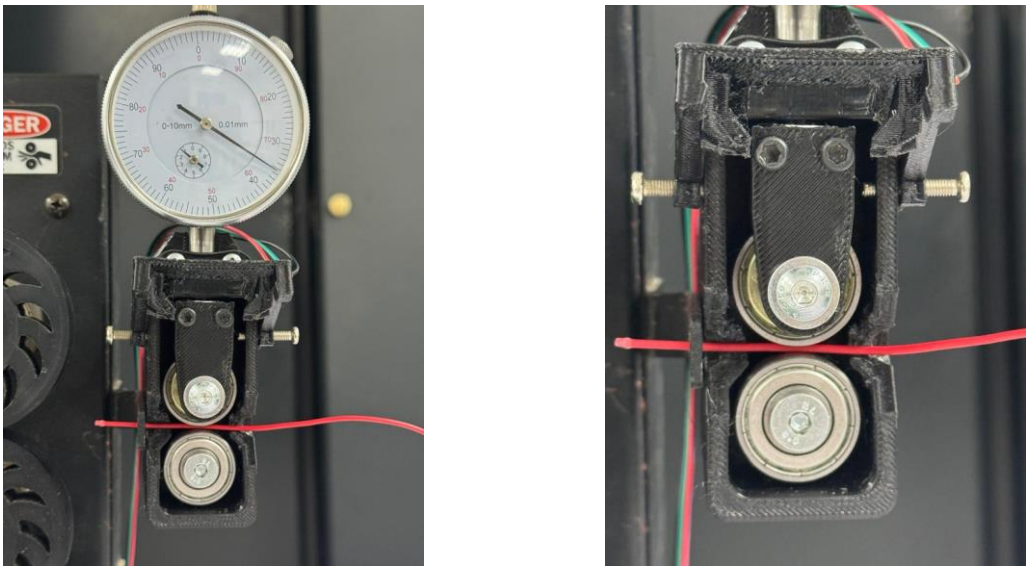
Figure 1: Electronic design of the sensing board in Proteus. (a) Circuit schematic: the SS49E Hall sensor (VCC, GND, OUTPUT) feeds its output to channel A0 of the ADS1115, whose SCL and SDA lines, together with the 3.3 V supply and ground, are routed to a four-pin header (PIPins) leading to the Raspberry Pi; the ADDR pin is tied to ground. (b) The corresponding PCB layout, in which the ADS1115

header is placed along the top, the Pi connector (PIPins) and the sensor connector (sensor1) sit along the bottom, and the red and blue traces show the two copper layers.

The basic measurement chain implemented by the gauge, illustrated in Figure 2, is

Filament diameter change → Bearing displacement → Magnet displacement → Hall sensor voltage

Dimensional variations in the filament thus cause a proportional shift of the magnet relative to the Hall sensor, producing a corresponding change in the electrical output. This mechanism is a practical way of converting minute mechanical deviations into an electrical signal that an embedded control system can process. The gauge is positioned after the cooling stage because measuring the polymer too early could deform the still-soft filament through mechanical contact and corrupt the readings.



(a) Assembled gauge (b) Bearing assembly

Figure 2: The 3D-printed sensor housing (gauge). (a) The assembled gauge with the red filament running horizontally between the lower, fixed bearing and the upper, movable bearing; a dial indicator (0–10 mm range, 0.01 mm graduation) is mounted above the housing as a mechanical reference, and the thin wires lead to the Hall-sensor circuit. (b) Close-up of the bearing pair: a change in filament thickness lifts or lowers the movable bearing, and this motion is transferred to the magnet next to the Hall sensor.

III.4. Data Calibration and Mathematical Model

The raw analog voltage produced by the Hall-effect sensor does not directly represent the filament diameter in millimeters; it varies with the proximity of the magnet. To obtain a precise conversion, the sensor was interfaced with the ADC, and a physical calibration was carried out to establish the mathematical relationship between the digitized voltage and the actual diameter. This calibration is essential because the sensor output depends strongly on the exact geometry and alignment of the 3D-printed mechanism, the magnet, and the custom PCB.

During calibration, known filament diameters ranging from 0 to 3 mm were passed through the sensor mechanism, and the resulting voltage readings were logged in Excel (Figure 3). Plotting the sensor voltage against the measured diameter (Figure 4) revealed a smooth, monotonic but non-linear relationship. Over the full range, the output changes by only about 0.2 V (from roughly 1.89 V at 0 mm to 2.09 V at 3 mm), which corresponds to an average sensitivity of about 66 mV/mm and makes the high resolution of the ADS1115 essential.

Diameter(mm)	Sensor Voltage(V)
0	1.891
1.75	1.984
1.8	1.9876
1.1	1.945
0.98	1.938
1.21	1.951
1.4	1.96
1.5	1.965
1.51	1.969
1.38	1.961
1.46	1.965
1.82	1.987
1.94	1.994
2.19	2.015
3	2.088
2.5	2.04
2.3	2.025
2.6	2.049
2.7	2.059
2.8	2.068
2.9	2.078
1	1.937

Figure 3: Excerpt of the calibration data logged in Excel. Each row pairs a known filament diameter (left column, in mm) with the sensor voltage measured at that diameter (right column, in V). The readings cover the 0–3 mm range, with the voltage rising from about 1.89 V at 0 mm to about 2.09 V at 3 mm.

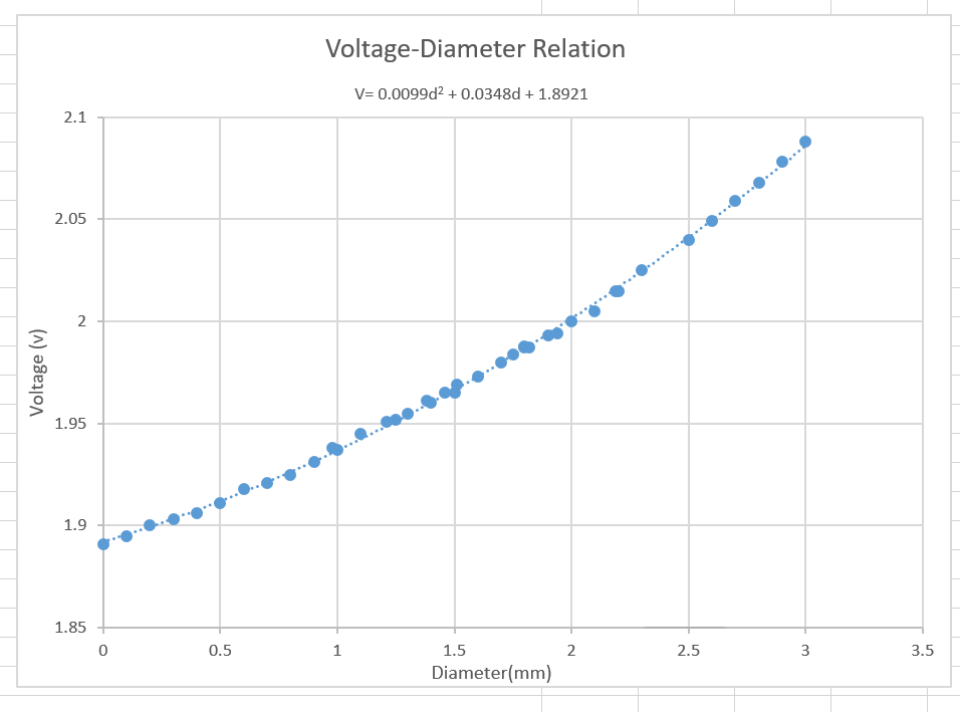


Figure 4: Voltage–diameter calibration curve. The blue markers are the measured sensor voltages for diameters from 0 to 3 mm, and the dotted line is the second-order polynomial trendline fitted to them, whose equation is printed above the plot area. The curve is slightly convex and strictly increasing, so every voltage corresponds to exactly one diameter.

A second-order polynomial trendline was fitted to the data, yielding the characteristic equation

$$V = 0.0099d^2 + 0.0348d + 1.8921 \quad (1)$$

where V is the sensor voltage in volts and d is the filament diameter in millimeters. Since the derivative $dV/dd = 0.0198d + 0.0348$ is positive throughout the calibrated range (between about 35 mV/mm at 0 mm and 94 mV/mm at 3 mm), the relationship is invertible. In the software, which reads V and must compute d, Equation (1) is rearranged into the standard quadratic form

$$ad^2 + bd + c = 0 \quad (2)$$

with coefficients $a = 0.0099$, $b = 0.0348$, and $c = 1.8921 - V$. Because a and b are positive and $c \leq 0$ for all voltages above the 0 mm level, the physically meaningful solution is the positive root given by the quadratic formula,

$$d = \frac{-b + \sqrt{b^2 - 4ac}}{2a} \quad (3)$$

which the control software evaluates in real time. If the discriminant is negative or the computed diameter is below zero, the software clamps the reading to 0 mm.

III.5 Closed-Loop Control Concept

Once a reliable measurement had been established, the next phase was to use the real-time data to regulate the extrusion process. An open-loop configuration would apply a fixed pulling-motor speed and could not correct the dimensional deviations caused by inherent process variations. The implemented system instead uses active closed-loop feedback.

The operator specifies the desired filament diameter (the reference, or set point) through a custom GUI on the Raspberry Pi 5, while the actual diameter is simultaneously obtained from the sensing hardware and the calibration equation. The difference between the target and the measured value is the control error,

$$e(t) = D_{ref} - D_{meas} \quad (4)$$

where D_{ref} is the desired diameter entered by the operator, D_{meas} is the measured diameter, and $e(t)$ is the control error, which is supplied continuously to the control algorithm. Because increasing the pulling speed thins the filament, the controller is implemented with the corresponding sign convention: a measured diameter above the target (negative e) raises the motor speed, and a diameter below the target lowers it.

III.6. PID Controller

The core control strategy was a Proportional–Integral–Derivative (PID) controller, implemented as a continuous background loop within the Python control script on the Raspberry Pi 5. It combines three standard terms:

- **Proportional term:** responds to the instantaneous error, $P = K_p e(t)$. A larger dimensional discrepancy produces a larger immediate corrective action.
- **Integral term:** accounts for the accumulated error over time, $I = K_i \int e(\tau) d\tau$. It is crucial for eliminating persistent steady-state errors in the filament thickness.
- **Derivative term:** evaluates the rate of change of the error, $D = K_d \, de(t)/dt$. It damps the response and prevents excessive overshoot during rapid dimensional changes.
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The complete control signal is the sum of the three terms:

$$u(t) = K_p e(t) + K_i \int_0^t e(\tau) d\tau + K_d \frac{de(t)}{dt} \quad (5)$$

The control script also incorporates dead-time compensation to account for the physical delay between the extrusion point and the sensor: the transport delay is estimated from the nozzle-to-sensor distance and the current line speed, and the correction is computed relative to the pulling speed that actually produced the filament now passing through the gauge. Additional safeguards include exponential smoothing of the sensor reading, a deadband around the target matched to the sensor resolution, integral

clamping with anti-windup, speed limits, and a limit on the change in speed per control cycle so that the motor never receives abrupt steps. The key parameters of the implementation are listed in Table 2.

Table 2: Main parameters of the control software running on the Raspberry Pi 5.

Parameter	Value
Proportional gain, K_p	12.0
Integral gain, K_i	0.4
Derivative gain, K_d	3.0
Control loop period	0.5 s
Pulling-motor speed limits	10 – 130 RPM
Maximum speed change per cycle	1.5 RPM
Low-pass filter coefficient	0.4
Error deadband (half of the 0.1 mm sensor resolution)	± 0.05 mm
Integral term clamp	± 15
Nozzle-to-sensor distance (for dead time)	1000 mm
Line speed per RPM	0.8 mm/s

The computed output sets the pull speed of the extruder by driving an L298N motor driver with a PWM signal. The overall control loop is therefore

Desired diameter → Python PID algorithm → L298N driver → Pulling motor → Extrusion process → Sensor / ADC → Calibration → Feedback

III.7. Effect of Pulling Motor Speed

Understanding the direct relationship between pulling speed and the final filament diameter was vital for tuning the control system. The pulling motor determines the rate at which the extruded polymer is drawn away from the nozzle. By modulating this speed through the L298N driver, the controller can stretch the material or allow it to accumulate.

For example, if the calculated diameter exceeds the reference value, the PID controller accelerates the pulling motor to thin the filament. The primary objective is to guide the measured diameter back to the set point without inducing aggressive oscillations or unstable mechanical behavior. The actual physical response is strongly multivariable, depending on thermal conditions, extrusion rate, and polymer rheology. This complexity highlights the multidisciplinary nature of the project, which required the integration of custom PCB hardware, Python control software, and mechanical extrusion dynamics.

III.8. Research and Experiments

A significant portion of the internship was devoted to empirical research and electronic hardware testing. I investigated the use of Hall-effect sensors in precision measurement, focusing on optimizing signal stability through the custom PCB layout and the integration of a high-resolution ADC. The calibration phase required careful experimentation and data modeling in Excel to derive the second-order polynomial transfer function.

Another major area of research was feedback control. Implementing the continuous background PID loop required an in-depth study of tuning methodologies and dead-time compensation, so that the L298N driver could respond to minute diameter fluctuations without inducing physical instability. The

internship therefore provided a rigorous platform for combining electronics theory, embedded software engineering, and practical hardware deployment.

IV. Internship Evaluation

This internship expanded my technical skill set and successfully bridged the gap between theoretical knowledge and practical engineering. Throughout the project, I gained critical hands-on skills in PCB design, hardware–software integration, mechanical design, data calibration, and Python programming, specifically in deploying PID control algorithms and developing graphical user interfaces.

The internship met my expectations: I had anticipated learning how the Filabot extruder works, and the opportunity to build a fully operational closed-loop control system from scratch strengthened my confidence in my problem-solving abilities. This experience will influence my future career plans, encouraging me to pursue a path in embedded systems engineering, circuit design, and control systems.

The activities I carried out were closely related to my university studies and allowed me to apply control theory to a physical industrial process. The courses most directly related to the internship were Control Systems, Digital System Design, Electronic Circuits, Programming, and Microcontrollers Lab.

V. Conclusion

The internship at the GUC provided a valuable opportunity to apply the theoretical knowledge acquired during my Electronics Engineering studies to a practical industrial application. The core project was the design and implementation of a closed-loop measurement and control system for 3D-printing filament extrusion. By combining custom PCB routing, a 3D-printed mechanical gauge designed in Autodesk Fusion, Hall-effect magnetic sensing, and a Python-based PID control algorithm, the system monitors the filament diameter in real time and dynamically regulates the pulling motor.

One of the most significant conclusions of this work is that accurate, closed-loop material regulation can be achieved economically, without expensive optical or laser equipment, when different engineering disciplines are tightly integrated. The reliability of the final system depended not only on the Python controller but also on the precision of the electronic hardware, the placement of the sensor, the signal calibration, and the mechanical CAD design. The project underscored that robust embedded electronic systems cannot be designed in isolation from the physical mechanisms they govern.

The internship also gave me a deeper perspective on the role of automation in the rapidly evolving 3D-printing industry. As material quality and process automation receive increasing attention, it became clear that dependable 3D prints are frequently limited by material inconsistencies, particularly variations in filament thickness. Dynamic feedback systems such as the one implemented here offer an effective solution by monitoring these variations and adjusting motor speed to keep filament tolerances within strict limits. A possible future improvement is the addition of an optical filament sensor to cross-check the Hall-effect measurement.

Overall, this internship strengthened my practical engineering skills and deepened my interest in embedded electronics, sensor integration, and software control. It demonstrated how custom electronics, software algorithms, and mechanical design can be unified to produce reliable solutions to industrial manufacturing problems, bridging my academic coursework and real-world engineering challenges.